E-Commerce SQL Assignment

Database Schema

The following SQL statements create the database schema and insert sample data.

```
CREATE TABLE Customers (
  customer_id INT PRIMARY KEY,
  name VARCHAR(50),
  city VARCHAR(50)
);

INSERT INTO Customers VALUES
(1, 'Aarav', 'Mumbai'),
(2, 'Riya', 'Delhi'),
(3, 'Kabir', 'Pune'),
(4, 'Simran', 'Chennai'),
(5, 'Anaya', 'Delhi');

CREATE TABLE Products (
  product_id INT PRIMARY KEY,
  product_name VARCHAR(50),
  category_id INT,
  price DECIMAL(10,2)
);

INSERT INTO Products VALUES
(101, 'Laptop', 1, 60000),
(102, 'Smartphone', 1, 30000),
(103, 'Headphones', 2, 2000),
(104, 'Shoes', 3, 2500),
(105, 'Washing Machine', 4, 40000);

CREATE TABLE Categories (
  category_id INT PRIMARY KEY,
  category_name VARCHAR(50)
);

INSERT INTO Categories VALUES
(1, 'Electronics'),
(2, 'Accessories'),
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(3, 'Fashion'),
(4, 'Home Appliances'),
(5, 'Furniture');

CREATE TABLE Orders (
  order_id INT PRIMARY KEY,
  customer_id INT,
  order_date DATE
);

INSERT INTO Orders VALUES
(201, 1, '2025-09-01'),
(202, 2, '2025-09-03'),
(203, 1, '2025-09-05'),
(204, 3, '2025-09-07'),
(205, 5, '2025-09-07');

CREATE TABLE OrderDetails (
  orderdetail_id INT PRIMARY KEY,
  order_id INT,
  product_id INT,
  quantity INT
);

INSERT INTO OrderDetails VALUES
(301, 201, 101, 1),
(302, 201, 103, 2),
(303, 202, 104, 1),
(304, 203, 102, 1),
(305, 204, 105, 1),
(306, 205, 103, 3);
```

SQL Assignment Questions

Answer the following queries using SQL based on the given schema. Each question is written in detail so that you can clearly understand what is expected.

1. Write a query to display all customers who have placed an order, along with their order IDs and the dates of those orders.

2. Write a query to show all customers and their corresponding orders. If a customer has not placed any order, still display their name with NULL for order details.

3. Write a query to display all products along with the category name to which each product belongs.

4. Write a query to display all categories along with the products in each category. If a category has no products, display the category name with NULL for product name.

5. Write a query to display each order along with the product names and quantities purchased in that order.

6. Write a query to display customer names along with the products they purchased and the quantity of each product.

7. Write a query to display the names of customers who have not placed any order.

8. Write a query to display the distinct category names of products that customers have purchased.

9. Write a query to display each order ID and the total amount spent in that order (total = price * quantity).

10. Write a query to find customers who live in the same city and display their names together (self join).

11. Write a query to display the names of customers who have purchased the product 'Headphones'.

12. Write a query to display customer names along with the product category they purchased.

13. Write a query to display the names of customers who purchased more than one product in a single order.

14. Write a query to generate all possible combinations of customers and products (cross join).

15. Write a query to display the names of categories in which no product has ever been ordered by any customer.

16. Write a query to display the names of customers whose total purchase value across all orders is greater than 50,000.

17. Write a query to display the number of orders placed by each customer.

18. Write a query to display the products purchased by customers who live in 'Delhi'.

19. Write a query to find the maximum quantity of a single product purchased in one order.

20. Write a query to display the names of customers who have purchased at least one product from the 'Electronics' category.

21. Write a query to display the names of categories where more than one product has been sold.

22. Write a query to display the order IDs of all orders that contain more than two different products.

23. Write a query to display the total revenue generated from each product category.